MUHAMMAD HANAN

Full Stack Developer | Machine Learning & AI

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SUMMARY

Full Stack Developer with hands-on experience building responsive, user-centric web applications using Python, Django, HTML, CSS, C++, and SQL (MySQL), along with working knowledge of Machine Learning and Deep Learning using scikit-learn and PyTorch. Skilled in scalable backend systems, relational databases, and predictive modeling with NumPy, Pandas, and Matplotlib. Proficient in Git and GitHub. Known for strong problem-solving, analytical thinking, and effective collaboration in team environments.

EDUCATION

Virtual University of Pakistan — **2025 – Continue**
BS Data Science — *Lahore, Pakistan*

Govt. Islamia Graduate College, Civil Lines, Lahore — **2022 – 2024**
Intermediate — *Lahore, Pakistan*

Laurelbank Public School — **2020 – 2022**
Matriculation — *Lahore, Pakistan*

PROJECTS

Chest X-Ray Pneumonia Screening — Deep Learning API & Web App | github ↗ — **2026**

- Built a chest X-ray pneumonia screening tool in PyTorch (EfficientNet-B0 transfer learning, class-weighted loss, early stopping), evaluated with sensitivity, specificity and AUC instead of accuracy alone.
- Added Grad-CAM heatmaps and served predictions through a FastAPI REST API and a Bootstrap 5 web interface.

Flood Prediction — Machine Learning Project | github ↗ — **2026**

- Built a regression model in Python to predict flood probability from 20 environmental and human-made factors using the Kaggle Playground S4E5 dataset (~1.1M rows).
- Compared Linear Regression, Random Forest, and XGBoost using scikit-learn pipelines; the best model (Linear Regression) achieved an R^2 of 0.85 on a held-out test set.
- Performed EDA and feature importance analysis, and deployed the trained model as an interactive Streamlit web app for live predictions.

Hanni's Espresso — Coffee Shop Web Application | live ↗ | github ↗ — **2026**

- Built a full-stack Order-Ahead web application using Python and Django, allowing users to browse interactive coffee menus and place direct orders.
- Designed a clean, responsive interface using HTML5 and CSS3 featuring dynamic item pricing and bill calculation logic.
- Engineered custom Django ORM models and database management deployed live on PythonAnywhere for real-time order processing.

EduTrack — Student Management System | live ↗ | github ↗ — **2026**

- Developed an administrative portal using Python and Django for course management, academic tracking, and real-time student record updates.
- Implemented Django Authentication and Role-Based Access Control (RBAC) to secure administrative routes and protect academic data.
- Designed interactive dashboard views styled with Bootstrap and backed by dynamic relational database CRUD operations.

SKILLS

Programming Languages:	Python, C++, SQL
Frontend Development:	HTML, CSS, Bootstrap
Backend Development:	Python, Django, REST APIs, File Handling
Machine Learning & AI:	Machine Learning, Deep Learning, PyTorch, scikit-learn, XGBoost, Streamlit
Data Science & Libraries:	Pandas, NumPy, Matplotlib
Databases:	MySQL, SQLite
Developer Tools & Platforms:	Git/GitHub, Postman, PythonAnywhere, Render

LANGUAGES

Spoken:	English (Fluent), Urdu (Native)

REFERENCES

Shall be furnished on demand